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T.E. (Computer Engineering)

WEB TECHNOLOGY

(2019 Pattern) (Semester - II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain the following:

[8]

- i) Process of transforming XML document.
 - ii) HTTP session
- b) What is Servlet? Explain the life cycle of servlet. Illustrate with example. **[9]**

OR

Q2) a) Compare doGet and doPost methods in servlet.

[9]

- b) Explain XML with respect to structure, declaration syntax, namespace. **[8]**

Q3) a) Write advantages of JSP over servlet and explain lifecycle of JSP.

[8]

- b) Explain the Struts architecture with neat diagram and also explain the benefits of Struts. **[9]**

OR

Q4) a) Write a JSP program to demonstrate use of page directive, Scriptlet Expression and Comment.

[9]

- b) Write the benefits of Web services and explain SOAP, Rest and UDDI. **[8]**

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Q5) a) Explain the following with respects to PHP. **[9]**

i) Arrays

ii) Function

iii) Control statements in PHP

b) How does this array work in PHP? Explain with example. **[9]**

OR

Q6) a) Explain object oriented way to connect MYSQL database with PHP. **[9]**

b) Draw and explain .NET framework with CLR, CLI. **[9]**

Q7) a) Explain scalar types and their operations in Ruby. **[9]**

b) What are the positive aspects of Rails, explain with example. **[9]**

OR

Q8) a) Write short note on: **[9]**

i) Rails with AJAX

ii) WAP and WML

b) What is EJB? Explain types of EJBs. **[9]**

→ → →

Solutions:

Q1 (a) Explain the following:

i) Process of Transforming XML Document ii) HTTP Session [8]

Ans:

i) Process of Transforming XML Document

Transforming an XML document means **converting it into another format**, such as another XML, HTML, or text.

This is usually done using **XSLT (Extensible Stylesheet Language Transformations)**.

1. What is XML Transformation?

- It is the process of changing the structure or presentation of an XML document.
- We use **XSLT** to apply rules and generate the required output.

2. Why is transformation needed?

- To convert XML into **HTML webpages**
- To convert XML into **another XML format**
- To extract selected data
- To generate **text reports, PDFs, or formatted output**

3. Components used in XML Transformation

1. **Source XML** → Input data
2. **XSLT Stylesheet** → Contains rules, templates
3. **XSLT Processor** → Executes transformation (e.g., SAXON)
4. **Output Document** → HTML/XML/Text/PDF

4. Steps in XML Transformation

Step 1: Create XML Document

Contains the actual data.

Step 2: Create XSLT File

Defines:

- How to select nodes
- How to display or modify data

Step 3: Use XSLT Processor

Processor reads XML + XSLT and applies matching templates.

Step 4: Generate Output

Output can be:

- HTML webpage
- Another XML file
- Plain text

5. Example Flow

XML → + XSLT → (Processor) → HTML Output

ii) HTTP Session

HTTP is **stateless**, meaning every request is treated separately.

A **session** allows the server to remember a user **across multiple requests**.

1. Definition

- An HTTP session is a **server-side storage** that keeps information about a user across multiple page requests.

2. Why session is needed?

Because HTTP cannot track:

- Login information
- Shopping cart data
- User preferences

Sessions store these temporarily.

3. How session works

1. User sends request → Server creates **Session ID**.
2. Session ID is stored in:
 - Cookie, or
 - URL rewriting
3. Browser sends Session ID with every request.
4. Server retrieves stored data for the user.

4. Data stored in session

- Username
- Email
- Cart items
- Temporary preferences

5. Session lifetime

- Created when the user visits
- Destroyed on:
 - Logout
 - Timeout
 - Closing browser (in some cases)

6. Session API in Java Servlet

```
HttpSession session = request.getSession();  
session.setAttribute("name", "Vijay");
```

Q1 (b) What is Servlet? Explain the life cycle of servlet. Illustrate with example.[9]

Ans:

A **Servlet** is a Java program that runs on a web server and generates **dynamic web content**. It handles requests and responses using **HTTP protocol**.

Servlet – Explanation

1. Definition

- Servlet is a **server-side Java component**
- Used to create **dynamic webpages**
- It extends the functionality of a web server

2. Features of Servlet

- Platform-independent
- Faster than CGI
- Supports sessions, cookies
- Secure and efficient

Servlet Life Cycle

The Servlet life cycle includes **5 main steps**:

1. Loading the Servlet

- Servlet container (like Tomcat) loads the servlet class.

2. Creating an Instance

- Only **one** instance of servlet is created.

3. init() Method

- Called once when servlet is created.
- Used for:
 - Initializing resources
 - Opening DB connections

```
public void init() {  
    // initialization code  
}
```

4. service() Method

- Handles every request.
- Calls:
 - doGet() for GET requests
 - doPost() for POST requests

```
public void service() {  
    // handles all requests  
}
```

5. destroy() Method

- Called before servlet is removed from memory.
- Used to:
 - Close DB connections
 - Free resources

```
public void destroy() {  
    // cleanup code  
}
```

Example Servlet

```
import java.io.*;
import javax.servlet.*;

public class HelloServlet extends GenericServlet {
    public void init() {
        System.out.println("Servlet Initialized");
    }

    public void service(ServletRequest req, ServletResponse res)
    throws ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<h2>Hello from Servlet</h2>");
    }

    public void destroy() {
        System.out.println("Servlet Destroyed");
    }
}
```

OR

Q2 (a) Compare doGet and doPost methods in Servlet.[9]

Ans:

doGet() and doPost() are two methods of the Servlet used to handle **HTTP GET and POST** requests.

Both are part of the HttpServlet class.

Example Code

doGet()

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    response.getWriter().println("This is GET request");
}
```

doPost()

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    response.getWriter().println("This is POST request");
}
```

Comparison (Minimum 9–10 points)

doGet()	doPost()
1. Handles HTTP GET requests.	1. Handles HTTP POST requests.
2. Sends data through URL as query string.	2. Sends data in HTTP body (not visible in URL).
3. Limited data transfer (because URL length is limited).	3. Can send large data (form data, files, hidden values).
4. Less secure because data appears in URL.	4. More secure because data is not visible in URL.
5. Mostly used for retrieving information.	5. Used for submitting information.
6. Can be bookmarked.	6. Cannot be bookmarked.
7. GET requests are idempotent (same result repeated).	7. POST requests are non-idempotent (submissions).
8. Faster because of simple request mechanism.	8. Slightly slower because it handles large/secure data.
9. Used to display pages, fetch data, simple queries.	9. Used for login forms, payments, uploading files.

Q2 (b) Explain XML with respect to structure, declaration syntax, namespace.[8]

Ans:

XML (Extensible Markup Language) is used to store and transport data. It is human-readable and machine-readable.

1. XML Structure

Point-to-point:

1. XML document contains user-defined **tags**.
2. There must be only **one root element**.
3. Elements must be **properly nested**.
4. All start tags must have **closing tags**.
5. Attribute values must be **in quotes**.
6. XML is **case sensitive**.

Example

```
<student>
  <name>Vijay</name>
  <roll>42</roll>
</student>
```

2. XML Declaration Syntax

Point-to-point:

1. XML documents often begin with **declaration tag**.
2. It tells XML version and encoding type.
3. It is optional but recommended.

Syntax

```
<?xml version="1.0" encoding="UTF-8"?>
```

Attributes

- **version** → version of XML
- **encoding** → character set (UTF-8, UTF-16)
- **standalone** (optional) → yes/no depending on external DTD

Example:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
```

3. XML Namespace

Concept

XML Namespace is used to **avoid name conflicts** when two XML documents have same tag names.

Why needed?

- Sometimes two XML files use same element names.
- Namespace uniquely identifies the element.

Syntax

Namespace is declared with **xmlns** attribute.

```
<book xmlns:h="http://www.w3.org/TR/html4/"
      xmlns:f="https://www.w3schools.com/furniture">
  <h:table>
    <h:tr>
      <h:td>Book Details</h:td>
    </h:tr>
  </h:table>
  <f:table>
    <f:name>Coffee Table</f:name>
  </f:table>
</book>
```

Point-to-point explanation:

1. xmlns:h is an HTML namespace.
2. xmlns:f is a furniture namespace.
3. Prefixes h: and f: identify which element belongs to which XML vocabulary.
4. Prevents **naming conflict** between <table> elements of different types.

Q3 a) Write advantages of JSP over Servlet and explain lifecycle of JSP.[8]

Ans:

1. Advantages of JSP over Servlets

1. JSP is more convenient for UI

- JSP is **HTML-based**, so creating UI pages is easy.
- Servlets mix HTML inside Java code using `println()`, which becomes messy.

2. Separation of concerns

- JSP = **View layer (UI)**
- Servlet = **Controller/logic layer**
- Better structure → easier maintenance.

3. Implicit objects available

- JSP gives built-in objects like request, response, session, application, out etc.
- In servlets we must create objects manually.

4. Automatically converted into servlet

- JSP is internally converted to a servlet → compiled → executed.
- So developer writes less Java code.

5. Tag libraries support

- JSP supports JSTL, custom tags, expression language.
- Makes code cleaner and reduces Java inside HTML.

6. Better for rapid development

- JSP allows quick changes in UI without recompiling Java classes.

2. JSP Life Cycle

JSP lifecycle has **5 major stages**:

1 Translation Phase

- JSP file is **translated into a Servlet class** by the container.

2 Compilation Phase

- The generated servlet file is **compiled into .class** bytecode.

3 Class Loading

- The servlet .class file is **loaded into memory** by the classloader.

4 Object Instantiation

- Container **creates an instance** of the JSP's servlet.

5 Initialization (jspInit())

- Called only once when JSP is first loaded.
- Similar to init() method in servlet.

6 Request Processing (_jspService())

- Called **for every request**.
- Contains code converted from JSP scriptlet, expression etc.

7 Destruction (jspDestroy())

- Called when JSP is removed from memory (shutdown).
- Used to release resources.

JSP Life Cycle Summary

JSP File → Translation → Compilation → Loading → Instantiation → jspInit() → _jspService() → jspDestroy()

Q3 b) Explain Struts architecture with neat diagram and benefits of Struts.[9]

Ans:

1. Struts Architecture (MVC Framework)

Struts follows **MVC (Model-View-Controller)** architecture.

Components:

1View (JSP)

- Displays data to the user
- Uses Struts tags for UI

2Controller (ActionServlet)

- Front controller
- Receives all requests
- Finds correct Action class from struts.xml
- Sends result to View

3Model (JavaBeans / Business Logic)

- Handles database operations, validations, calculations

4struts.xml (Configuration File)

- Maps URL → Action class → Result page
- Defines interceptors, validations, global exceptions

5Action Class

- Contains actual logic
- Returns result string such as success, error

6Interceptors

- Pre-processing and post-processing
- Example: validation, authentication, logging

2. Flow / Diagram

Client Request



ActionServlet (Controller)



Reads struts.xml



Executes Interceptors



Calls Action Class (Model)



Returns result string



Selects JSP View



Response to Client

3. Benefits of Struts

1. Based on MVC pattern

- Clear separation of logic, design and data → easier maintenance.

2. Centralized Configuration

- struts.xml manages all URL mappings, validations, error pages.

3. Reusable components

- Interceptors, actions, models can be reused easily.

4. Built-in validation framework

- Reduces manual coding of validations.

OR — Q4 a) Write a JSP program to demonstrate page directive, Scriptlet, Expression and Comment.[9]

Ans:

Here is a **clean exam-ready JSP program**:

```
<%@ page language="java" contentType="text/html" %>  <!-- Page Directive -->
```

```
<html>
```

```
<body>
```

```
<%-- This is a JSP Comment (not sent to browser) --%>
```

```
<%
```

```
    // Scriptlet
```

```
    String name = "Vijay";
```

```
    int a = 10, b = 20;
```

```
    int sum = a + b;
```

```
%>
```

```
<h2>Using JSP Expression:</h2>
```

```
<p>Name: <%= name %></p>
```

```
<p>Sum of 10 and 20 is: <%= sum %></p>
```

```
</body>
```

```
</html>
```

Demonstrated Items:

- **Page Directive** → `<%@ page ... %>`
- **Scriptlet** → `<% Java code %>`
- **Expression** → `<%= value %>`
- **Comment** → `<%-- comment --%>`

Q4 b) Write the benefits of Web Services and explain SOAP, REST and UDDI.[8]

Ans:

1. Benefits of Web Services

1. Platform Independent

- Communication possible between Java, Python, .NET, PHP etc.

2. Language Independent

- Uses XML/JSON, so any language can process data.

3. Loosely coupled

- Services work independently; no need to know internal logic.

4. Interoperability

- Can be used across different systems, devices, OS.

5. Reusability

- Same service can be used by multiple applications.

6. Standardized

- Use open standards like HTTP, XML, JSON, SOAP.

7. Supports distributed systems

- Suitable for cloud, SOA architectures.

2. SOAP (Simple Object Access Protocol)

- SOAP is an **XML-based messaging protocol**.
- Uses **HTTP, SMTP** for communication.
- Very secure and reliable for enterprise apps.

SOAP Message Structure

```
<Envelope>  
  <Header>...</Header>  
  <Body>  
    <getInfo>  
      <id>101</id>  
    </getInfo>  
  </Body>  
</Envelope>
```

3. REST (Representational State Transfer)

- REST is an **architectural style** using **HTTP methods**: GET, POST, PUT, DELETE
- Uses **JSON or XML**
- Lightweight, faster, widely used in web/mobile apps.

REST Example

GET https://api.example.com/users/101

4. UDDI (Universal Description, Discovery and Integration)

- UDDI is a **registry/directory** of web services.
- It contains details like:
 - Service name
 - Methods
 - URLs
 - Company information

Purpose

- Allows clients to **find and discover** web services.
- Works like “Yellow Pages” for web services.

Q5) a) Explain the following with respect to PHP. [9]

Ans:

(i) Arrays in PHP

Concept in Brief

Arrays store multiple values in a single variable. PHP supports 3 types of arrays.

Points

1. Array is a collection of values under one name.
2. Values can be accessed using numeric or string keys.
3. PHP supports:
 - Indexed Arrays
 - Associative Arrays
 - Multidimensional Arrays
4. Arrays are flexible and can store different data types.
5. Arrays are created using `array()` or `[]`.

Example

```
$colors = ["Red", "Blue", "Green"]; // Indexed array  
echo $colors[1]; // Output: Blue
```

(ii) Functions in PHP

Concept in Brief

A function is a block of reusable code.

Points

1. Function reduces code duplication.
2. Function is declared using the function keyword.
3. Functions may accept parameters.
4. Functions may return values.
5. Built-in functions (like `strlen`) and user-defined functions are supported.

Example

```
function add($a, $b) {  
    return $a + $b;  
}  
echo add(5, 3); // Output: 8
```

(iii) Control Statements in PHP

Concept in Brief

Control statements decide the flow of execution.

Points

1. **Conditional statements:** if, else, elseif, switch
2. **Looping statements:** for, while, do-while, foreach
3. **Jump statements:** break, continue
4. Used to implement decisions and repetition in programs.

Example

```
$age = 20;  
if($age >= 18) {  
    echo "Eligible";  
} else {  
    echo "Not Eligible";  
}
```

Q5) b) How does this array work in PHP? Explain with example. [9]

Ans:

An array in PHP maps **keys** → **values**.
The key can be **numeric** or **string-based**.

How PHP Arrays Work – Pointwise

1. PHP arrays are **ordered maps**.
2. Each element has:
 - **Key** (index)
 - **Value**
3. Keys can be:
 - Automatic numeric (0,1,2...)
 - User-defined numeric
 - Strings

1. PHP arrays are dynamic (size grows automatically).
2. Arrays can store **any data type** including arrays and objects.
3. Access elements using \$array[key].
4. Arrays internally use **hash tables**, so access is fast.

Example

```
<?php
$student = [
    "name" => "Vijay",
    "age" => 22,
    "city" => "Pune"
];

echo $student["city"]; // Output: Pune
?>
```

OR-Q6) a) Explain object-oriented way to connect MySQL database with PHP. [9]

Ans:

PHP provides **OOP-based MySQL connection** using **MySQLi class** or **PDO class**.

Method 1: Using MySQLi OOP

Steps (Pointwise)

1. Create connection object
2. Check connection
3. Write query
4. Execute query
5. Fetch records
6. Close connection

Example Code (OOP MySQLi)

```
<?php
// Step 1: Create connection object
$conn = new mysqli("localhost", "root", "", "college");

// Step 2: Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Step 3: Query
$sql = "SELECT id, name FROM students";
$result = $conn->query($sql);

// Step 4: Display records
while($row = $result->fetch_assoc()) {
    echo $row["id"]." - ".$row["name"]."<br>";
}

// Step 5: Close connection
$conn->close();
?>
```

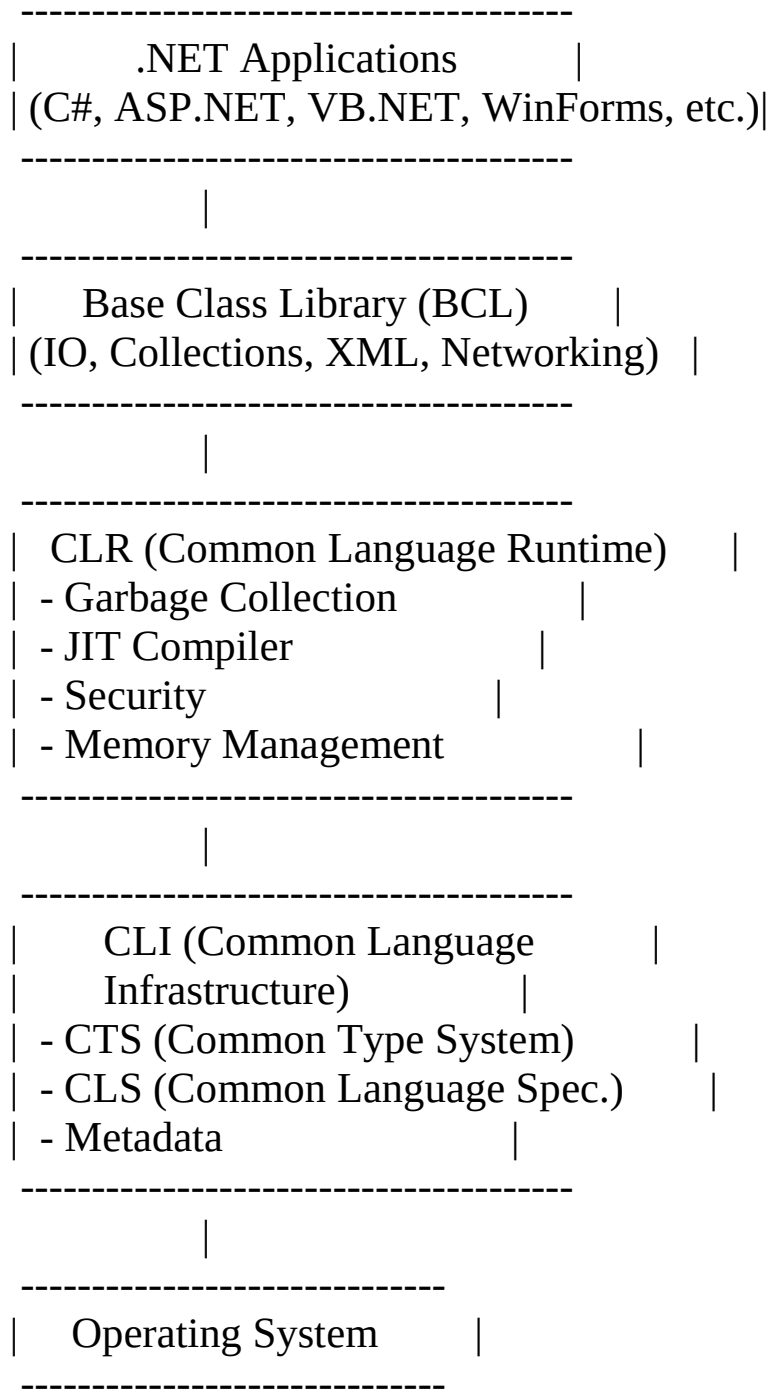
Q6) b) Draw and explain .NET Framework with CLR, CLI. [9]

Ans:

.NET Framework is a platform for building Windows and web applications. It has:

- **CLR (Common Language Runtime)**
- **CLI (Common Language Infrastructure)**
- **Base Class Library**

Diagram



Explanation

1) .NET Applications

- Programs developed using C#, VB.NET, ASP.NET.
- Executes on CLR.

2) Base Class Library (BCL)

- Ready-made classes for:
 - Strings
 - Files
 - Collections
 - Database
 - Networking

3) CLR (Common Language Runtime)

Execution engine of .NET

Responsibilities:

1. Converts IL → Machine code (via JIT compiler)
2. Handles memory management
3. Performs garbage collection
4. Manages exceptions
5. Provides security model

4) CLI (Common Language Infrastructure)

- International standard for .NET architecture.
- Components:
 - **CTS (Common Type System)** → defines data types
 - **CLS (Common Language Specification)** → ensures language interoperability
 - **Metadata** → information about assemblies

Q7 a) Explain scalar types and their operations in Ruby.[9]

Ans:

Ruby scalar types represent **single/simple values**. Main scalar types are:

1) Integer

- Stores whole numbers

```
age = 20
```

Operations:

```
10 + 5  # 15
```

```
10 - 3  # 7
```

```
10 * 2  # 20
```

```
10 / 2  # 5
```

```
10 % 3  # 1
```

2) Float

- Stores decimal values

```
price = 99.50
```

Operations:

```
5.5 + 2.3  # 7.8
```

3) String

- Sequence of characters

```
name = "Ruby"
```

String Operations:

```
"Hello " + "Ruby"    # "Hello Ruby"
```

```
"Hi" * 3              # "HiHiHi"
```

```
"Age: #{age}"        # String interpolation
```

4) Boolean

- Represents **true/false**

flag = true

Boolean Operations:

true && false # false

true || false # true

!true # false

5) Nil

- Represents “no value”

data = nil

Comparison Operations

Used for integers, floats, and strings:

a == b

a != b

a > b

a < b

"abc" == "abc"

Assignment Operations

x += 5 # x = x + 5

x -= 3

Q7 b) What are the positive aspects of Rails? Explain with example.[9]

Ans:

Ruby on Rails is a **web application framework** that focuses on **productivity, speed, and less code**.

Positive Aspects of Rails

1) MVC Architecture

Clear separation of:

- **Model** → Data
- **View** → UI
- **Controller** → Logic

This improves maintainability.

2) Convention over Configuration (CoC)

Rails reduces configuration work because it follows **standard naming conventions**.

Example:

A model named Student automatically maps to **students** table.

3) DRY (Don't Repeat Yourself)

Write code only once and reuse it across application.

Example:

Shared methods in helpers and partial views reduce repetition.

4) Built-in Database Support (ActiveRecord)

Rails automatically generates SQL queries.

Example:

```
@student = Student.new(name: "Vijay")
```

```
@student.save
```

No SQL queries are required manually.

5) Scaffolding

Rails can automatically generate:

- Model
- Controller
- Views
- Database migration

Command:

```
rails generate scaffold Student name:string age:integer
```

6) Security Features

Rails protects against:

- SQL Injection
- CSRF
- XSS

Automatically provided without extra code.

7) Rapid Development

Rails has many built-in helpers for forms, sessions, authentication, etc.

Example Form:

```
<%= form_with model: @student do |f| %>
  <%= f.text_field :name %>
  <%= f.submit "Save" %>
<% end %>
```

8) Strong Community and Gems

Thousands of ready-made components called **gems**, e.g.,

- Devise (Authentication)
- Kaminari (Pagination)

Example of Rails Simplicity

Controller

```
def index
  @students = Student.all
end
```

View

```
<% @students.each do |s| %>
  <p><%= s.name %></p>
<% end %>
```

Minimal code → complete dynamic page.

OR- Q8 a) Write short notes on: (i) Rails with AJAX (ii) WAP & WML)[9]

Ans:

i) Rails with AJAX (5 Marks)

AJAX = **A**synchronous **J**avaScript **A**nd **X**ML.

Rails supports AJAX natively to update only part of a webpage **without refreshing the entire page**.

How Rails supports AJAX

1) remote: true

Used in forms and links.

Example:

```
<%= form_with url: "/students", remote: true do |f| %>
  <%= f.text_field :name %>
  <%= f.submit %>
<% end %>
```

2) JavaScript Response File (.js.erb)

Rails sends JavaScript instead of HTML.

create.js.erb

```
alert("Student added successfully!");
```

Benefits

- Faster response
- Less bandwidth
- Better user experience
- Smooth UI updates without page reload

ii) WAP and WML (4 Marks)

WAP (Wireless Application Protocol)

- A protocol to access **internet on mobile devices** (old mobile phones).
- Optimized for low bandwidth and small screens.

Architecture Components

- WAE (Application Environment)
- WSP (Session Protocol)
- WTP (Transaction Protocol)
- WTLS (Security Layer)
- WDP (Datagram Protocol)

WML (Wireless Markup Language)

- Markup language used for WAP.
- Similar to HTML but lightweight.
- Pages are called **cards** inside a **deck**.

Example WML Code

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN">
<wml>
  <card id="card1" title="Hello">
    <p>Hello WML!</p>
  </card>
</wml>
```

Q8 b) What is EJB? Explain types of EJBs. (9 Marks)

Ans:

EJB (Enterprise JavaBeans) is a **server-side component** used to build **large, secure, scalable enterprise applications** in Java.

Types of EJB

1) Session Beans

Used to perform business logic.

i) Stateless Session Bean

- Does **not** maintain client state.
Example: banking interest calculation.

ii) Stateful Session Bean

- Maintains **session state** for each client.
Example: shopping cart.

iii) Singleton Session Bean

- Only **one instance** exists for entire application.
Example: cache, application-wide configuration.

2) Message-Driven Beans (MDB)

- Handles **asynchronous messages** (JMS).
- Does not return response to client.
Used in:
 - Notifications
 - Email services
 - Background tasks

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Best Regards, **Vishwajeet Londhe**